Mahidol University
Wisdom of the Land

WEEK 2: BAYES. THEOREM

Probabilistic Approach

BAYESIAN IDEA

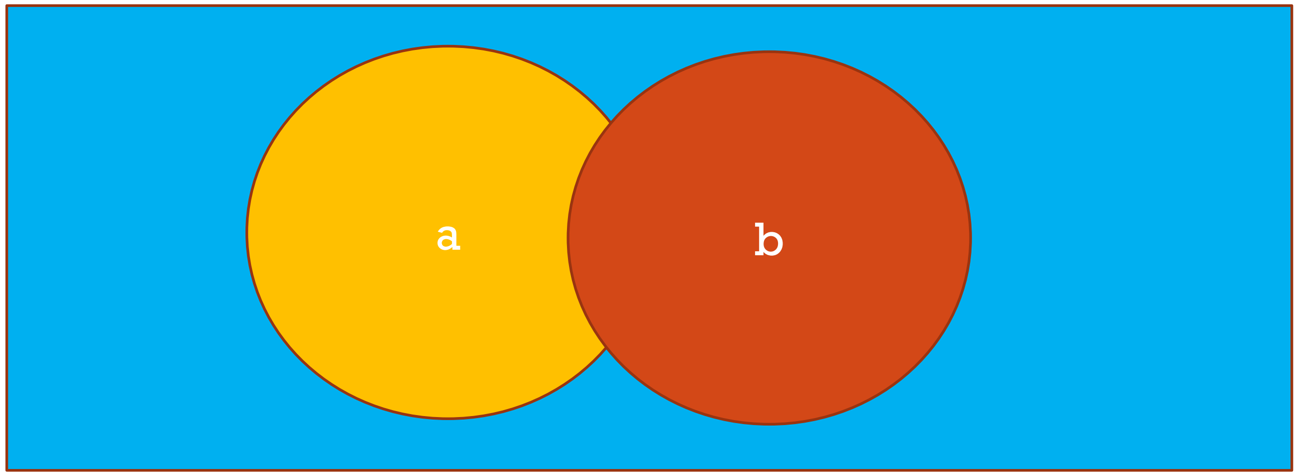
- Not all decision are certain
- Probabilistic estimation
- Probability is purely estimated from data
- Parameters is needed to be estimated

BASIC PROBABILITY

- p(a) the probability of an event, ranged between 0-1

- p(a,b) the probability of a joint ranged between 0-1.
- Normally

p(a,b)<=p(a) and p(a,b)<=p(b)



JOINT PROBABILITY

Assumed:

- A is raining
- B is T<30c

The event that both A, B is a joint event.
Therefore
P(A, B) □ It's both rain and T<30
less than
P(A) □ It's only rain
P(B) □ The temperature is only <30

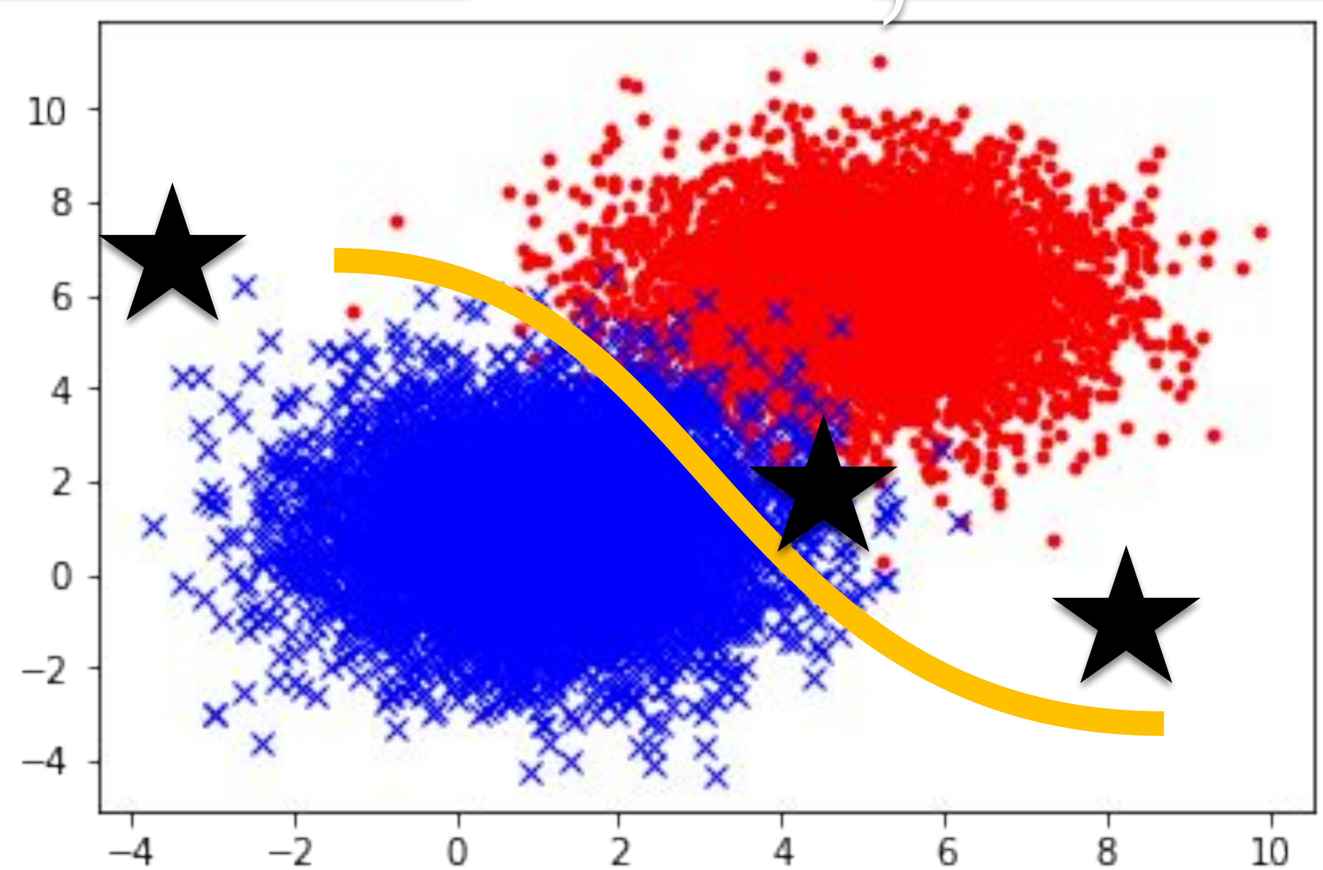
BAYES' THEOREM

- The existing probability is used to determine 'likelihood' of each object.
- The recognition step will calculate the **'posterior'**

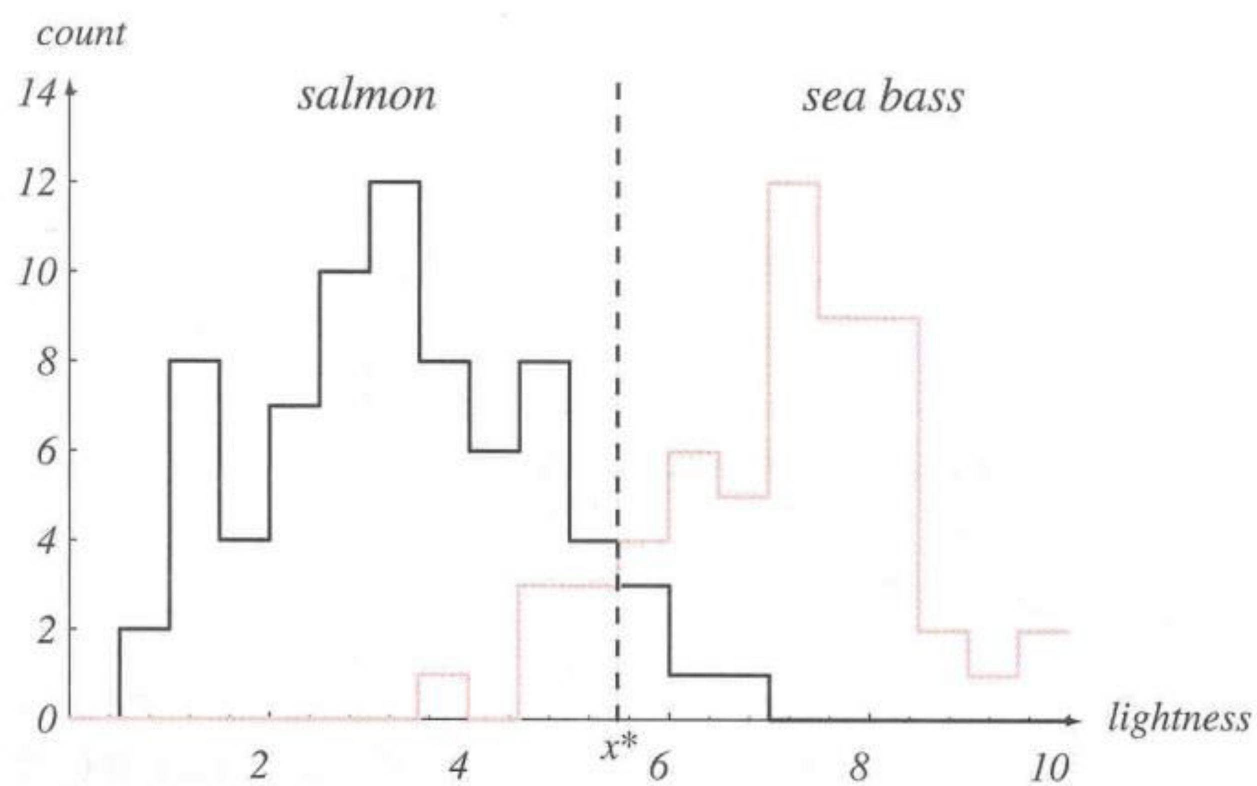
conditional probability

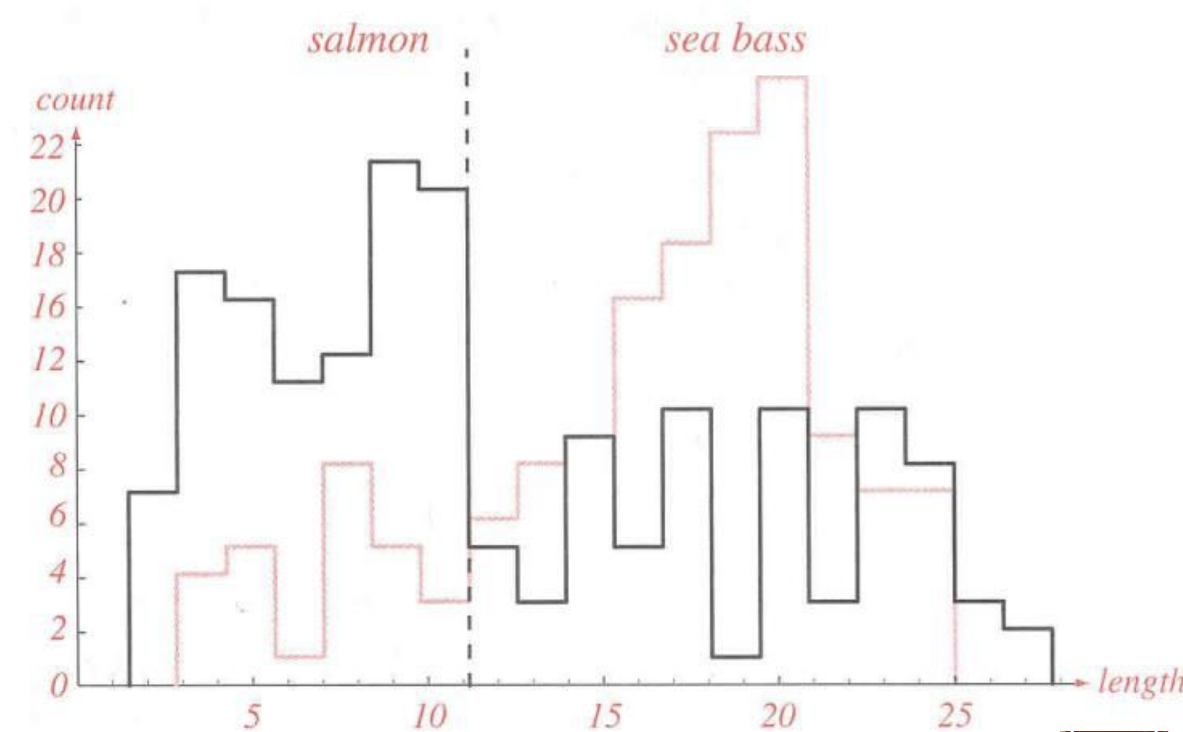
p(D|Salmon)

p(D|Seabass
)



AN EXAMPLE OF FISH CLASSIFICATION





CREATE PROBABILITY

Basic Concept

- What do we want, to find?

  p(Salmon|x) or p(Seabass|X)

- What do we (need to) have?
  - P(Salmon) P(Seabass)
  - P(x|Salmon) P(x|Seabass)

or

  - P(x, Salmon) P(x,Seabass)
  - P(x)

BAYES THEOREM

Posterior

likelihood

prior

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)}$$

Evidence

- **Prior** is a portion of each class or object type. **P(C)**
- **Likelihood**: From the given data, what the probability this of this data.
- **Evidence***:* How possible each data is?

MORE THEORY

Marginal Probability

$$\mathrm{P}(x) = \sum_x P(x, y)$$

Joint Probability

$$\begin{aligned} P(x, y) &= P(x|y)P(y) \\ &= P(y|x)P(x) \end{aligned}$$

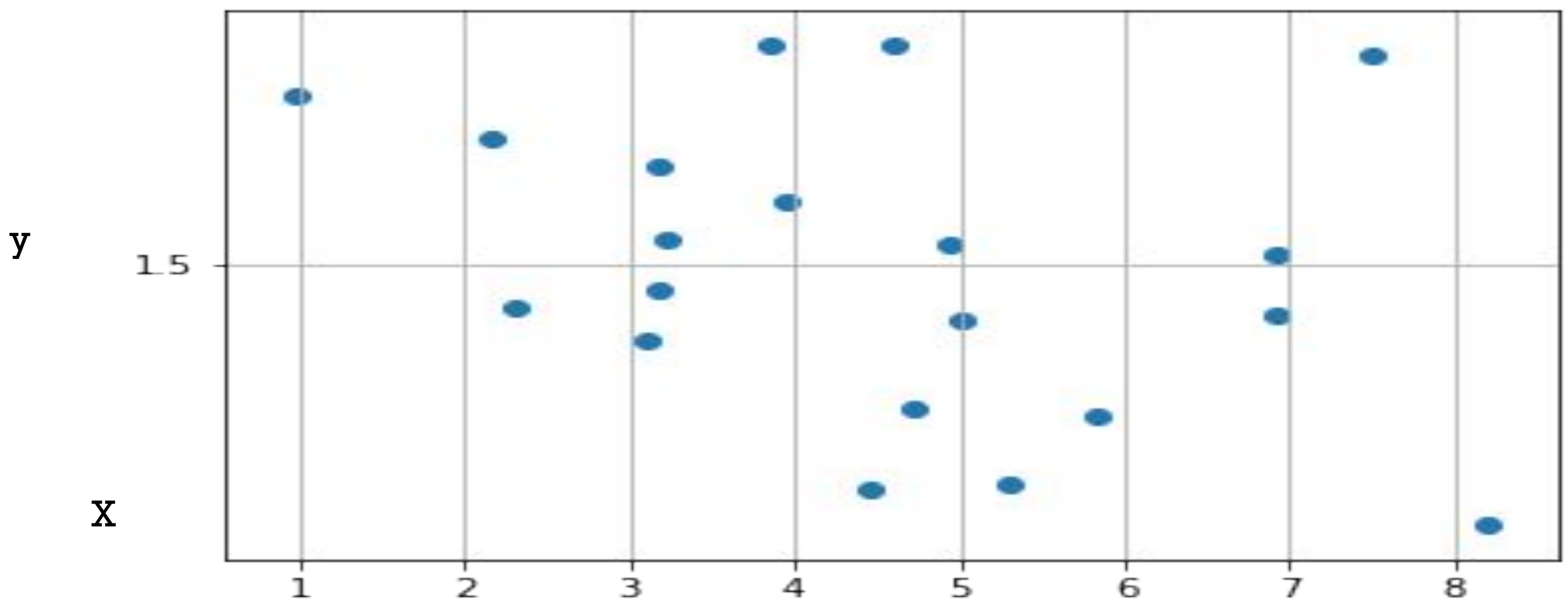
PUT THEM TOGETHER

$$\mathrm{P}(y|x) = \frac{\mathrm{P(x|y)P(y)}}{P(x)} = \frac{\mathrm{P(x,y)}}{P(x)}$$

$$\mathrm{P}(y|x) = \frac{\mathrm{P(x|y)P(y)}}{\sum_x P(x|y)p(y)}$$

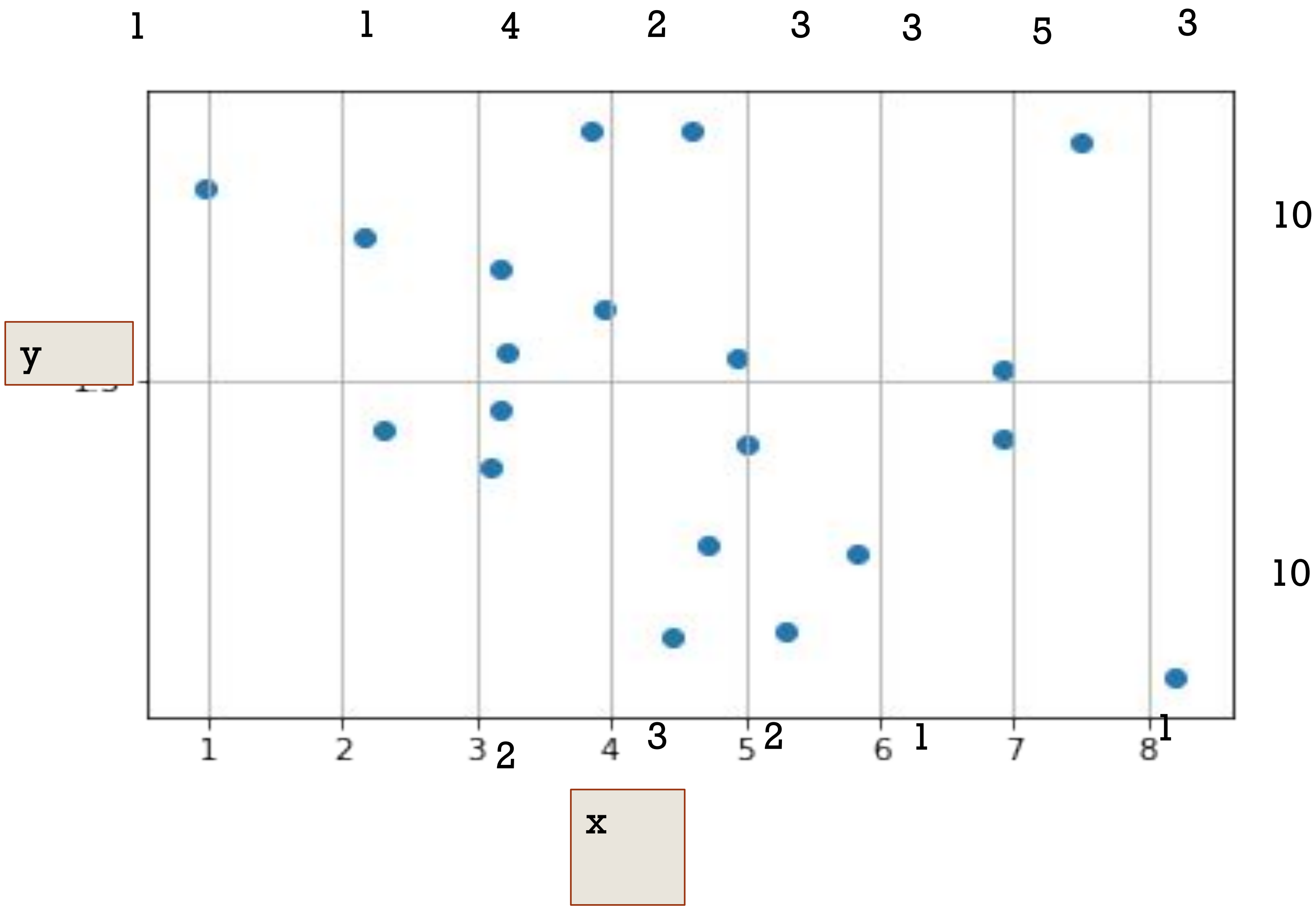
1-D DISTRIBUTION

Joint probability



x=	1	2	3	4	5	6	7	8	9
y=2	1	0	1	4	2	0	1	1	0
=1	0	0	1	2	3	2	1	0	1

EXTRACT INFORMATION



- Joint P(x=2,y=1)=1/20=5%

JOINT PROBABILITY

x=	1	2	3	4	5	6	7	8	9	SUM
y=2	1	0	1	4	2	0	1	1	0	10
y=1	0	0	1	2	3	2	1	0	1	10

$P(y=2, x=1) = 1/20$

$P(y=1, x=4) = ?/20$

$P(y=2, x=9) = ?/20$

MARGINAL PROBA

Marginal Probability

$$\mathrm{P}(x) = \sum_x P(x,y)$$

x=	1	2	3	4	5	6	7	8	9	SUM
y=2	1	0	1	4	2	0	1	1	0	10
y=1	0	0	1	2	3	2	1	0	1	10

P(y =1) = (0+0+1+2+3+2+1+1) /20 =10/20

P(y =2) = ? /20

P(x=1) = 1/20 + 0/20 =1/20

P(x=2) = ?

MARGINAL PROBA

Marginal Probability

$$P(x) = \sum_x P(x,y)$$

x=	1	2	3	4	5	6	7	8	9	SUM
y=2	1	0	1	4	2	0	1	1	0	10
y=1	0	0	1	2	3	2	1	0	1	10

P(y =1) = (0+0+1+2+3+2+1+1) /20 =10/20

P(y =2) = ? /20

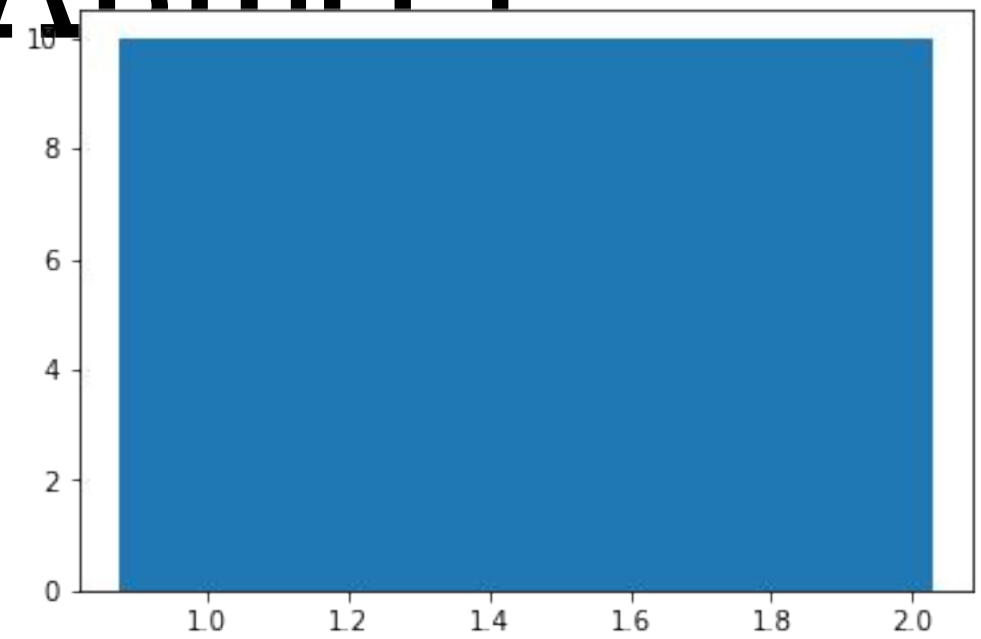
P(x=1) = 1/20 + 0/20 =1/20

P(x=2) = ?

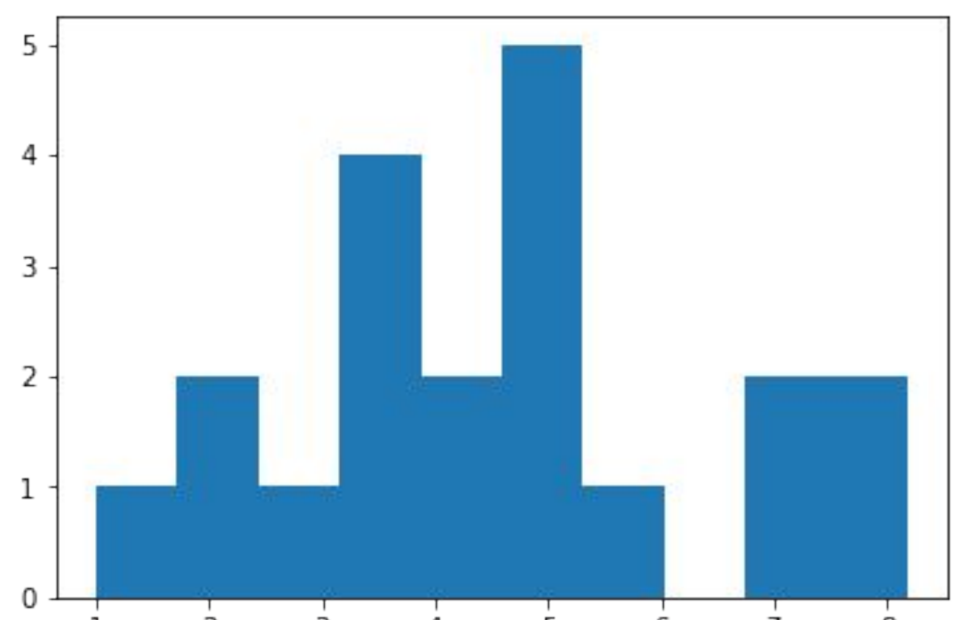
MARGINAL PROBABILITY

P(y)	SUM
y=2	10/20
y=1	10/20

P(y)



P(x)	1	2	3	4	5	6	7	8	9
	1/20	0/20	2/20	6/20	5/20	2/20	2/20	1/20	0/20



P(x)

CONDITIONAL PROBABILITY

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)} = \frac{P(x,y)}{P(x)}$$

x=	1	2	3	4	5	6	7	8	9	SUM
y=2	1	0	1	4	2	0	1	1	0	10
y=1	0	0	1	2	3	2	1	0	1	10

- $P(y=1|x=1)=p(y=1,x=1)/p(x=1) =0/1 =0$
- $P(y=1|x=5)=p(y=1,x=5)/p(x=5) =3/5 =0.6$
- $P(x=1|y=1)=?$
- $P(x=5|y=2)=?$

EXAMPLE1: FRUITY EXAMPLE

- Box 1 contains 8 apples and 4 oranges
- Box 2 contains 10 apples and 2 oranges
- Boxes are chosen with equal probability
- What is the probability of choosing an apple?
- If apple is chosen, what is the probability that it come from box 1?

SOLUTION

- $P(Box1)=P(Box2)=0.5$
- $P(Apple|Box1)= 8/12$
- $P(Apple|Box2)= 10/12$
- $P(Apple)=P(Apple|Box1)P(Box1) + P(Apple|Box2)P(Box2)$

$=8/12*0.5+10/12*.5 =0.75$

- $P(Box1|Apple)=?$

EXAMPLE 2:MEDICAL EXAMPLE

- Determining the false positive values
- If a tested patient has the disease the test returns a positive result 99% of the time or with probability 0.99
- If a tested patient does not have the disease the test returns a positive result 5% of the time or with probability 0.05
- What is the probability of the wrong positive test result? 5%?

wikipedia

CALCULATION

- Suppose 0.1% of the population has that disease
- Let A represent the condition of having the disease
- B represent positive test result
- The prob. of having the disease given the positive test result is

$$P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|notA)P(notA)}$$
$$= \frac{0.99\text{x}0.001}{0.99\text{x}0.001 + 0.05\text{x}0.999}$$

- Probability of of having the disease if the test is positive =0.019
- Probability of not having disease if the test is positive

false positive

= 1− 0.019 = 0.98 or 98%

QUESTION???

- How do you think it can be apply with the **iris** data?
- How do you think it can be apply with the **Image** data?

CLASSIFICATION

$$P(label|features) = \frac{\text{P(features|lable)P(label)}}{\sum_x P(\text{features,}label)}$$

$$P(label|features) = \frac{\text{P(features|lable)P(label)}}{\sum_x P(\text{features|}label)\text{P(label)}}$$

MAXIMUM LIKELIHOOD THEOREM

- $\mathrm{P}(label|features) = \frac{\mathrm{P(features|label)P(label)}}{\sum_x P(\mathrm{features}|label)\mathrm{P(label)}}$

- $\mathrm{P}(label|features) \propto \mathrm{P(features|label)P(label)}$

- Label_answer= argmax(P(features|lable)P(label))

N - DIMENSIONAL TECHNIQUES

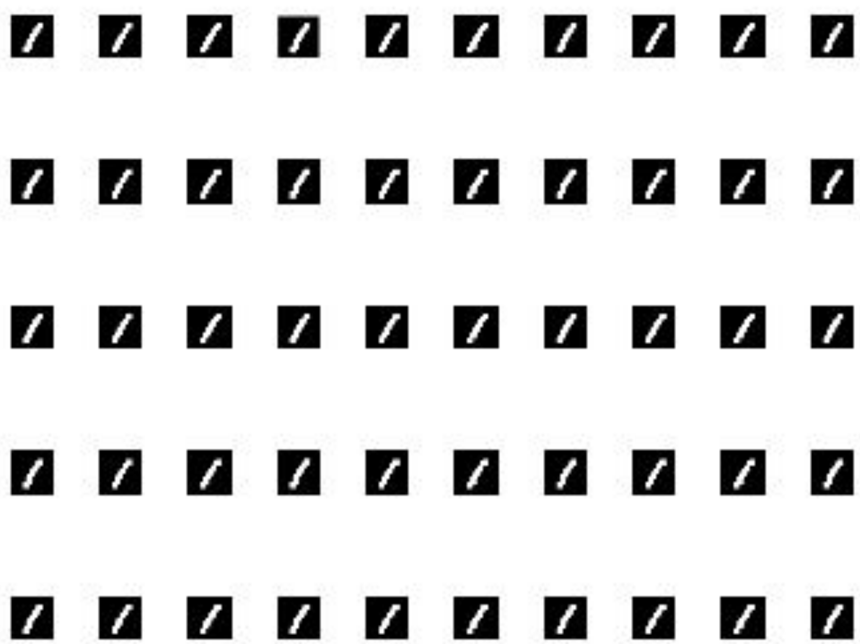
- Bayes' Bayes assumes Independence

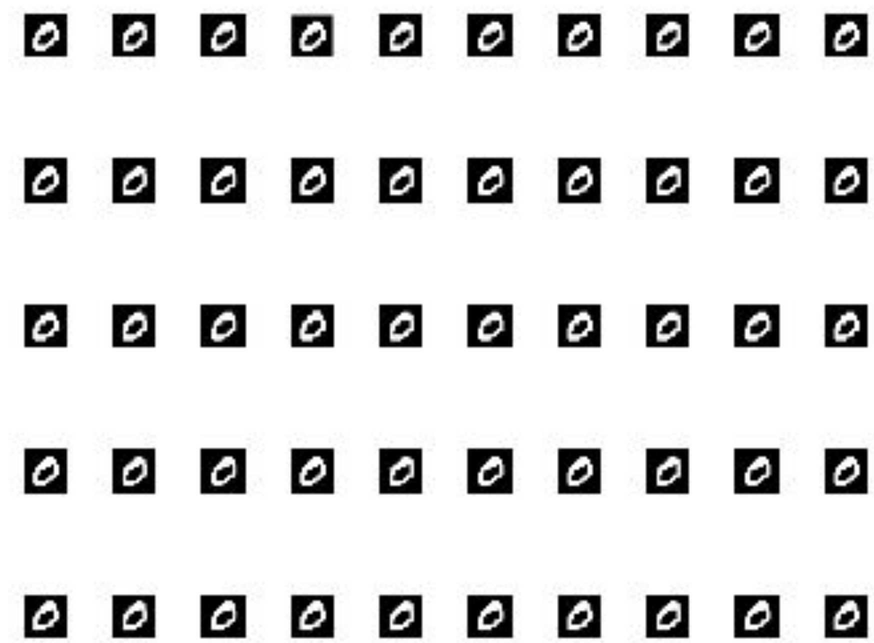
$$p(c|f1,\ldots,fn) \propto p(c)p(f1|c)\ldots p(fn|c)$$

- Example

$$\mathbf{P(digit1|px1,px2,px3) \propto P(digit1)\ (px1|digit1)\ P(px2|digit1)\ P(px3|digit1)}$$

HANDWRITTEN RECOGNITION





- How to apply for 28x28 pixel handwritten recognition?

NOTE ON MUTUALLY EXCLUSIVE EVENT

- p(a,b)=p(b|a)p(a)

When it is independent p(b|a)=p(b)

Dependent event example

Tossing two coins

what is the probability of getting head and head?

Independent event example

What is the probability of both sunny, raining?

TRY WITH SIMPLE BAYES

- How to use Bayes theorem with pictures?

p(0|D)=p(0|P1=1,P2=1,P3=1) = ?

$$p(0 \mid D) = p(0)p(P1, P2, P3)$$

Assume each pixel is independent

(the occurrence of one pixel is not dependant on another i.e. cannot find p(p1|p2))

$$p(0 \mid D) = p(0)\prod p(P_i)$$

BAYES THEOREM ON DIGIT RECOGNITION

1) Find prior P(0), p(1)
2) Find likelihood

$$p(P1, P2, P3, \ldots., Pn \mid 0) = p(P1 \mid 0)\, p(P2 \mid 0)\, p(P3 \mid 0) \ldots p(Pn \mid 0)$$

3) Get the posterior

$$p(0 \mid P1, P2, P3, \ldots., Pn) = \frac{prior * likelihood}{p(P1, P2, P3, \ldots, Pn)}$$

$$p(0 \mid P1, P2, P3, \ldots., Pn) \propto \frac{prior * likelihood}{p(P1, P2, P3, \ldots, Pn)}$$

Can omit even more with uniform prior

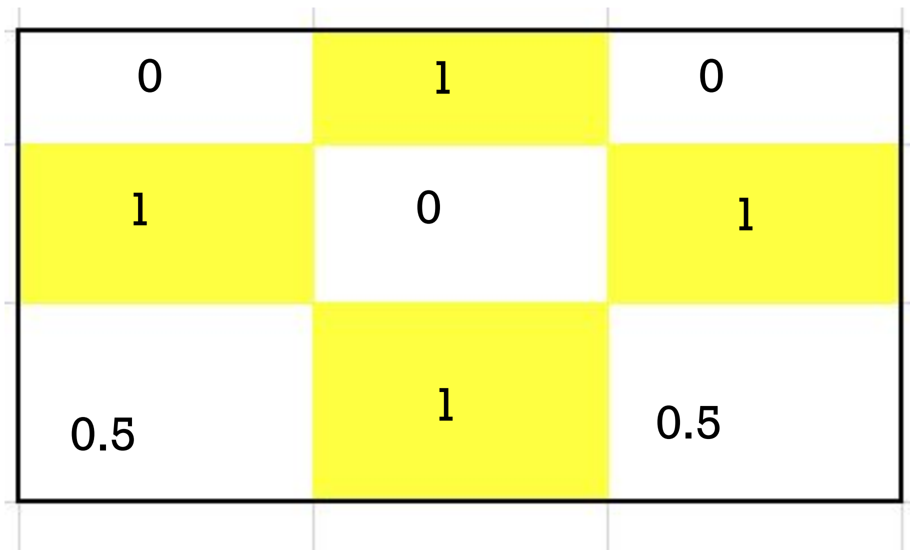
EXAMPLE

Number 0

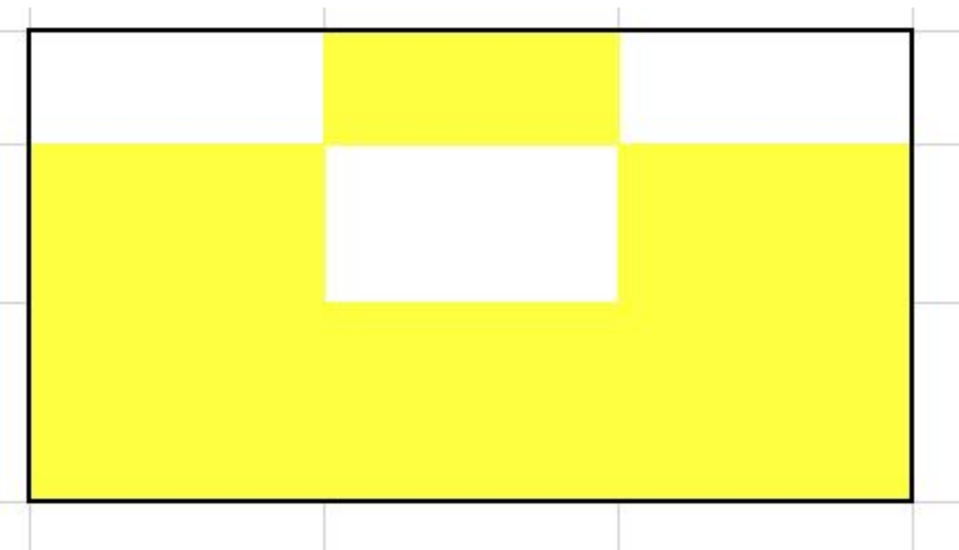
Number 1

P(PX=YELLOW|NUMBER)

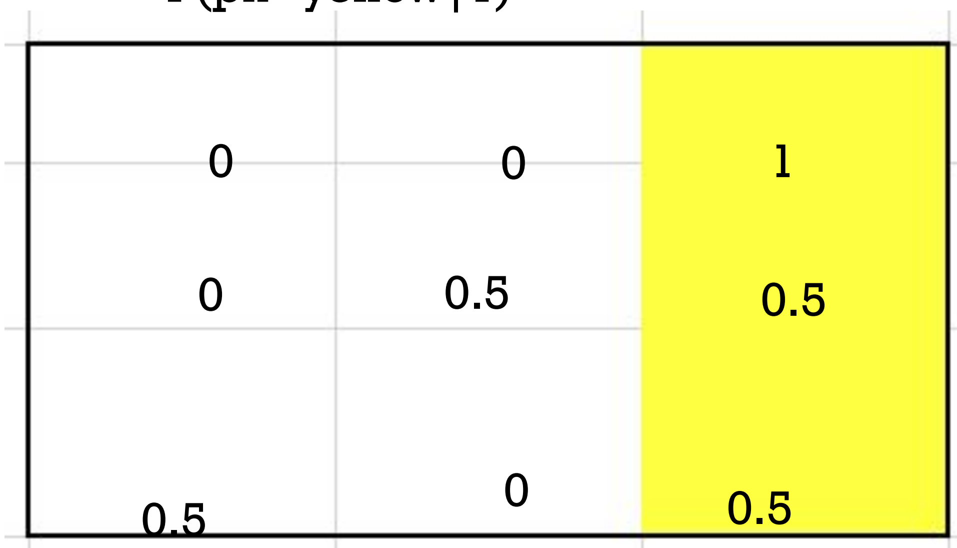
P(px=1|0)



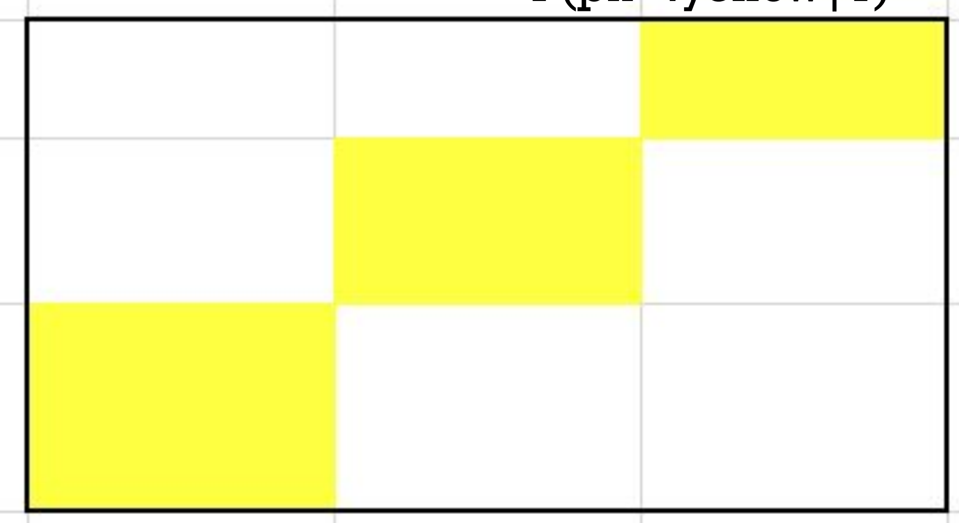
P(px=!yellow|0)



P(px=yellow|1)



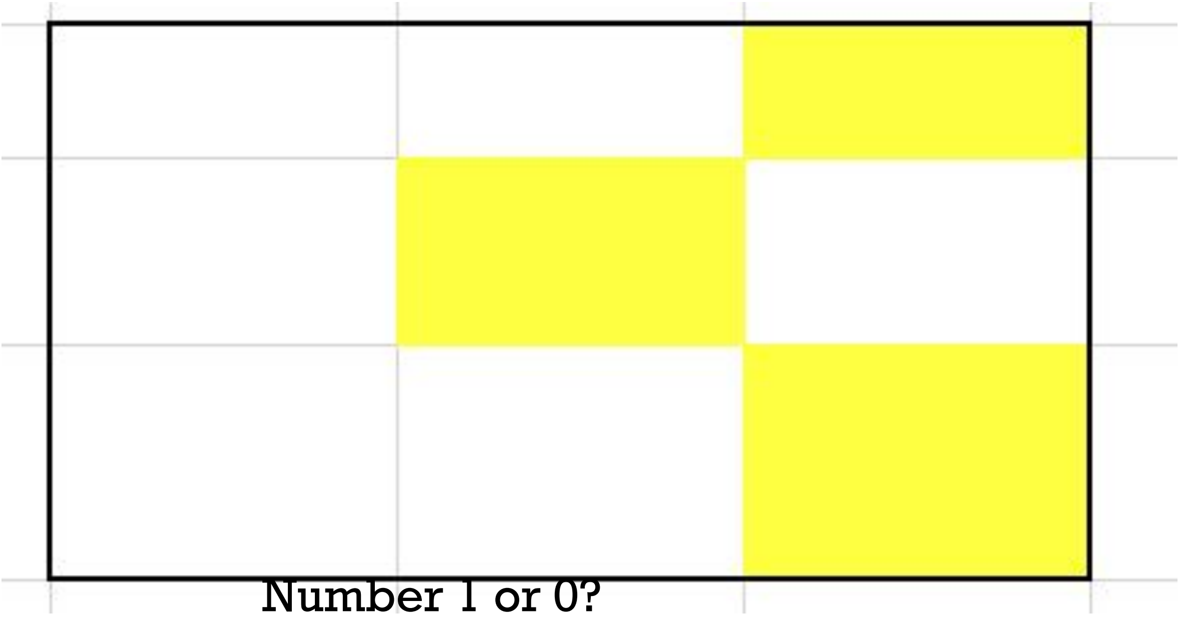
P(px=!yellow|1)



FIND THE PROBABILITY

$$p(P1, P2, P3, \ldots, Pn \mid 0) = p(P1 \mid 0)\, p(P2 \mid 0)\, p(P3 \mid 0) \ldots p(Pn \mid 0)$$

- P(0) / P(1)
- P(D|0) = 1*0*0*0*1*0*0*0.5*0
- P(D|1)=1*1*1*1*0.5*0.5*0.5*1*0.5=0.625
- P(0|D)= P(D|0) *P(0)/ P(D) = 0*0.5/(0*0.5+.625*0.5)
- P(1|D)=1



Number 1 or 0?

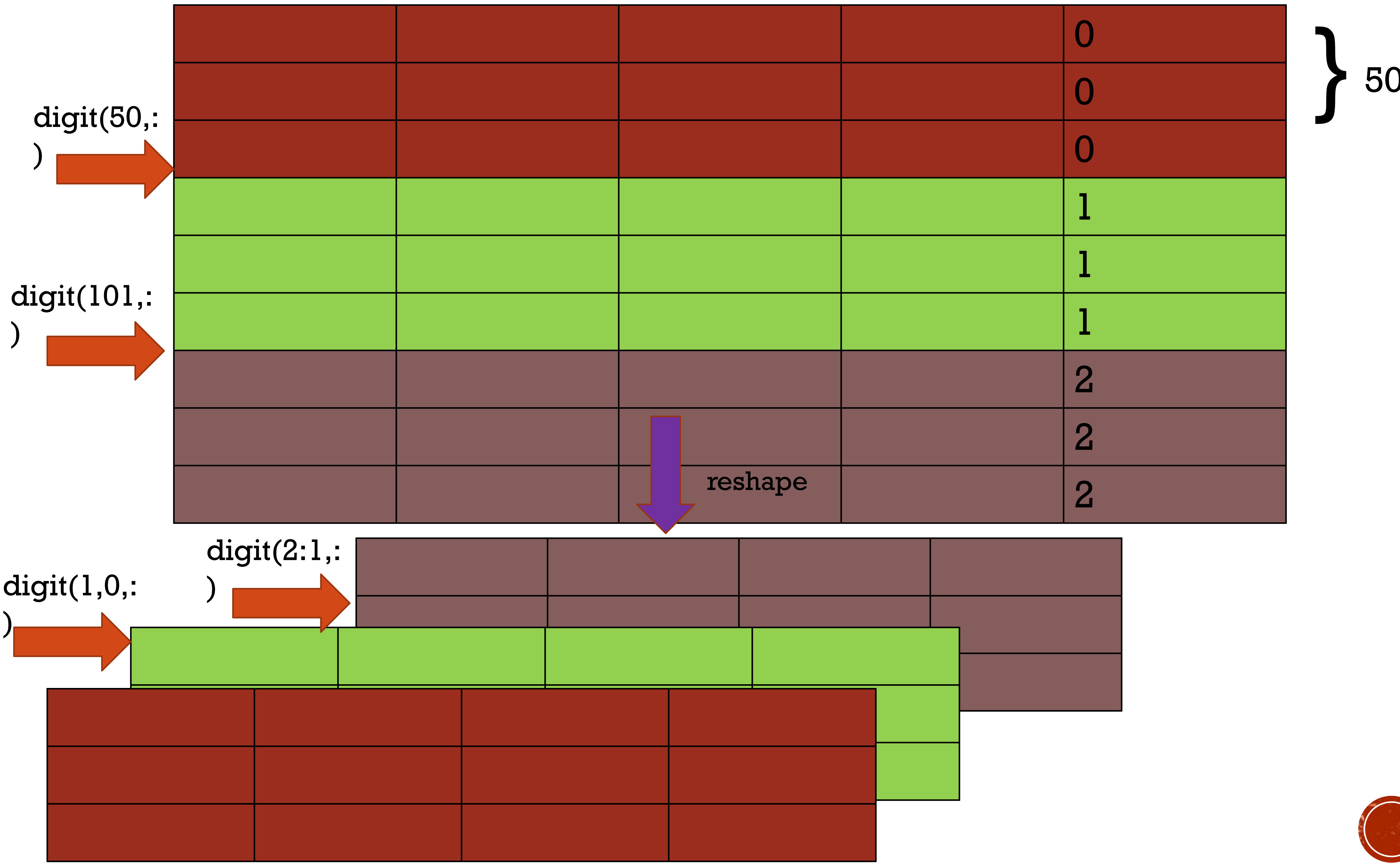
IMAGE

- 0 – 256 shades
- GreyScale -> binary(B&W)
- 0-127 □0
- 128-255 =>1

DEMO

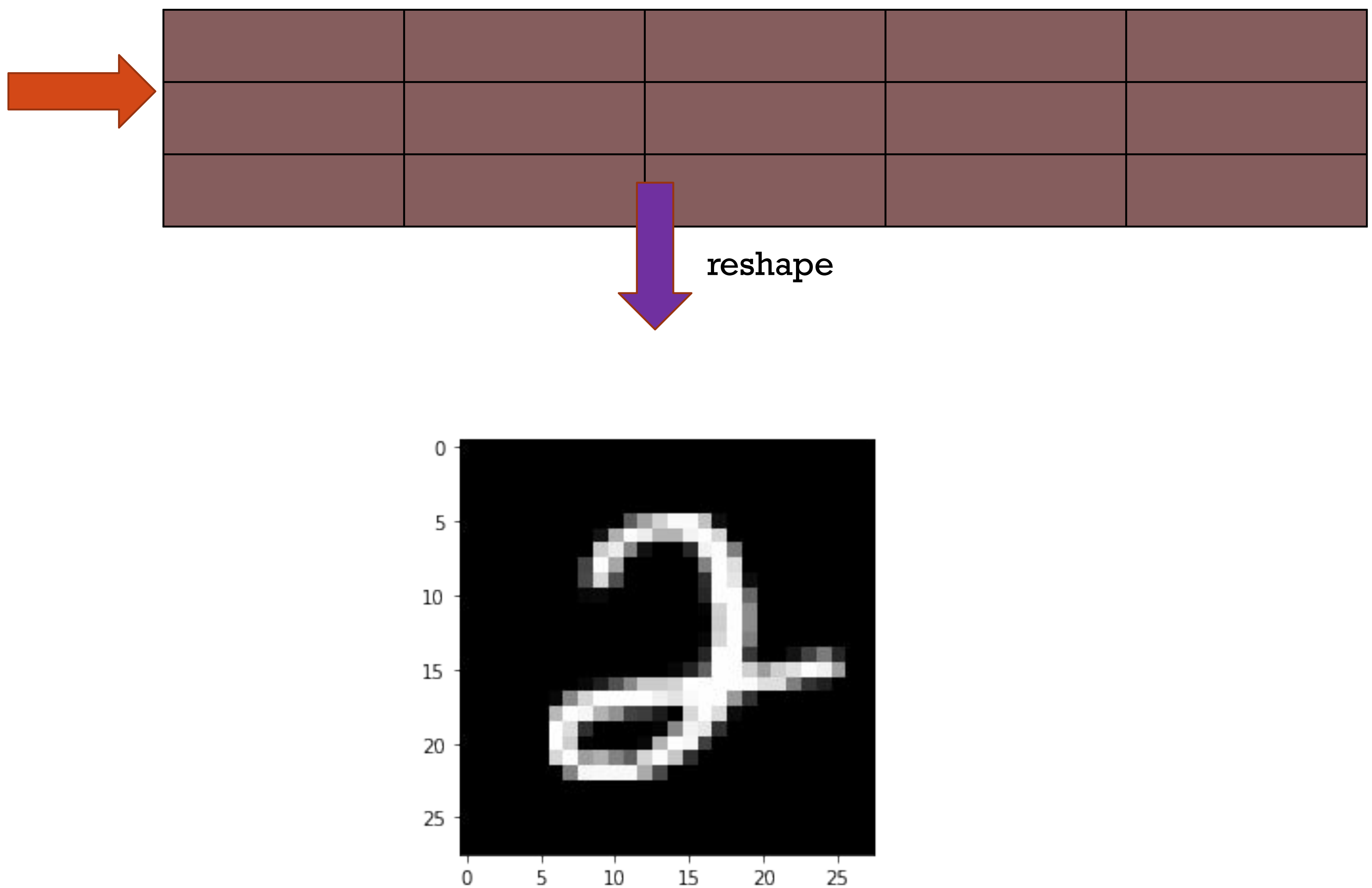
- Use digit_read (from csv)

1. RESHAPE THE DIGIT



1.1 VISUALIZED IT

digit(2:1,:)



2. BINARIZE THE VALUE

0	140	200	100
1	179	190	125
100	120	111	200

binarize
>128

0	0	1	0
0	1	1	0
0	0	0	1

3. CALCULATE LIKELIHOOD

1	1	1	1	0
0	1	1	1	0
1	1	0	1	0
1	0	1	1	1
0	1	1	1	1
1	0	1	0	1
				2
				2
				2

} 50

P(x=1 | digit 0)

0.33	1	0.67	1	0
0.67	0.33	1	0.67	1

P(x=1 | digit 1)

4. CALCULATE LIKELIHOOD(2)

P(x=1|digit 0)

0.33	1	0.67	1	0
0.67	0.33	1	0.67	1

P(x=1|digit 1)

P(x=0|digit0)=1-P(x=1|digit0)

0.67	0	0.33	0	1
0.33	0.67	0	0.33	0

P(x=0|digit1)=1-P(x=1|digit1)

5.CALCULATE THE POSTERIOR

calculated likelihood

0.67	0	0.33	0	1
0.33	0.67	0	0.33	0

New number?

1	0	1	0	1

p(digit|0)=0.2211

multiply

1*0.67	1*(1-0)	1*0.33	1*(1-0)	1

p(digit|1)=0

multiply

1*0.33	1*(1-0.67)	1*0	1*(1-0.33)	(1-0)

HOMEWORK I

- Download/load digit.csv From the website
- **Binarize** the image and use this theorem
- Try to recognize 3 types of numbers using Bayes' theorem